Galaba Vamsi

Mechatronics at IIT Bhilai. Robotics software, simulation, and controls.
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SUMMARY

Graduating May 2027. 15 merged pull requests to open-source robotics tools: HFlow (robot-data conversion), Inspect Robots (evaluation), and Cerulion (runtime). I wrote and maintain inspect-robots-wandb on PyPI. Simulation and controls work in ROS 2/Gazebo, MuJoCo, MATLAB/Simulink, and Unity. Co-author of the Inspect Robots workshop paper (CoRL 2026 SPAIS) and 2 RIS papers.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology Bhilai | **Coursework:** Optimal Control, Advanced Control Theory, Deep Learning for Computer Vision, AI and ML, Digital Image Processing, DSA, OOP. Aug 2023 – May 2027

OPEN SOURCE

HFlow Hebbian Robotics (YC S26) | Repository | My PRs Sep 2026 – Oct 2026

- Open-source SDK for verifying the quality of robot training data. 8 merged PRs, mostly making robot-log and LeRobot conversion keep data whole or refuse it with a clear reason, e.g. depth videos that would silently drop from 12-bit to 10-bit (#400).
- Benchmark PR (#367): timed three cold 1080p30 checks, reported no safe speedup, and left the filter graph unchanged.

Inspect Robots and inspect-robots-wandb Robocurve (YC S26) | Repository | PyPI | Plugin source Sep 2026 – Oct 2026

- Evaluation framework for physical-AI policies. 2 merged PRs: a configurable LLM retry policy across 5 wire protocols (#476) and unknown git state no longer logged as clean (#475). 3 open: #514 checkpoint/resume for long eval sets (in review), #550, #551.
- Wrote and maintain inspect-robots-wandb, a community W&B plugin on PyPI, after the maintainers asked for the sink to live outside the core: 100% coverage gate, CI on Python 3.10–3.13, trusted publishing.

Cerulion Open-source Rust robot runtime that records a run and re-executes it against new code | Repository Oct 2026

- 5 merged PRs: 1 CLI fix (#216, ros2 attach --dry-run outside a workspace) and 4 documentation/comment corrections.

EXPERIENCE

Research Intern IIT Bhilai | Advisor: Dr. Arzad Alam Kherani | RIS-SIM v2 | v1 May 2025 – Jul 2025

- Built RIS-SIM, an open-source Python emulator for RIS-assisted radio links in LoS/NLoS scenarios with USRP-style SDR interfaces; designed RIS unit cells in CST Studio Suite for RF energy harvesting (IEEE INDICON 2025, co-author).
- Rebuilt it in 2026 as RIS-SIM v2 (6 scenarios, 103 tests), then from scratch as sremu (Projects).

Founder, Robotropic Physical-AI startup | Website Oct 2026 – Present

- Building Harness, a workspace where a factory's team and AI agents change robot skills safely: each change checked against safety limits, rolled out cell by cell, and recorded.

Co-founder, Human Slop Anti-AI social app | Web app | Play Store 2026 – Present

- Co-building an Expo (React Native) app for web and Android with a Fastify API and PostgreSQL on Supabase; live on the web and Google Play (50+ downloads).

PROJECTS

Quarter-Car Active Suspension Study (course project, simulation only, 2026) Reproduced Na et al. (IEEE TSMC 2022) in MATLAB/Simulink: after fitting two plant parameters to its Case-10 experiment, the unchanged controller lands within about 6% of its error indices. The paper's Case-B gains fail a hold-out test, and LQR and skyhook match or beat it at equal effort. Includes a Cortex-M4 timing estimate. Site | GitHub

Whiteboard Cleaning Robot (2025, with Gajanand Kumawat) 3-DOF Cartesian gantry for a 90 × 90 cm whiteboard, simulated in ROS 2 Jazzy and Gazebo Harmonic: URDF/Xacro, ros2_control, 10 cm-overlap raster path, and logged coverage analysis. GitHub

Prompt Robot Arm (2026, simulation) MuJoCo: a prompt such as "sort red, blue, green" is parsed (rule-based for now) into a plan and a pick-and-place sequence, moving from a kinematic gripper to a SCARA arm with analytic inverse kinematics. GitHub

Pneumatic Sorting Digital Twin (offline simulation) (2026) Unity simulation of three electropneumatic training kits: simulated axes, logical PLC I/O, HMI screens, and telemetry record and replay. Not yet connected to a physical PLC. GitHub

sremu – Timed RIS Control Emulator (2026, MIT) IQ-level RIS emulator: timestamped RIS commands act on an exact, logged sample; deterministic and real-time controller-in-the-loop modes; runs unmodified GNU Radio 3.10; ~2,100 tests. GitHub

Arista Wi-Fi RRM (team; Inter IIT Tech Meet 14.0, 5th place, Dec 2025) Client-aware radio resource management in NS-3 with I/Q-based interference classification (STFT, DBSCAN, Dual-CNN; 16 classes) on a 5-AP, 50-client 802.11k/v topology. Slides

PUBLICATIONS

- **Co-author** — *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306, 2026; accepted to the SPAIS Workshop at CoRL 2026 (poster). arXiv
- **Co-author** — *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025.
- **Co-author** — *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*, Ad Hoc Networks (Elsevier), 2026.
- **Sole author (preprint draft, arXiv submission pending)** — *Timed RIS Control in the Loop: A Sample-Accurate IQ-Level Emulator with Characterized Timing*, 2026. Draft

SKILLS

Robotics, simulation, and controls: ROS 2 (rclpy, ros2_control, URDF/Xacro), Gazebo Harmonic, MuJoCo, Unity (C#), MATLAB/Simulink, LQR and Kalman filter design, LeRobot datasets, NS-3, GNU Radio
Languages: Python, C/C++, Rust (open-source contributions), TypeScript/JavaScript, SQL, GLSL
ML and scientific computing: PyTorch, OpenCV, NumPy, CuPy, Apache Arrow, Weights & Biases, MEEP (FDTD), CST Studio Suite
Tooling: Docker, Git, Linux, GitHub Actions, pytest, Ruff, mypy, PyPI trusted publishing, FastAPI, PostgreSQL, AWS

ACHIEVEMENTS

5th place, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team, Dec 2025) | **5th of 232 teams**, WEAR Dataset Challenge @HASCA 2026 (Kaggle, final private leaderboard) | **AIR 200**, Amazon ML Challenge 2026 (team) | **Top 1%**, JEE Advanced | **PMSS** scholarship (2024) | **Subroto Mukherjee** Scholarship (IAFBA, 2023) | **YC Startup School**, Bengaluru (attended)